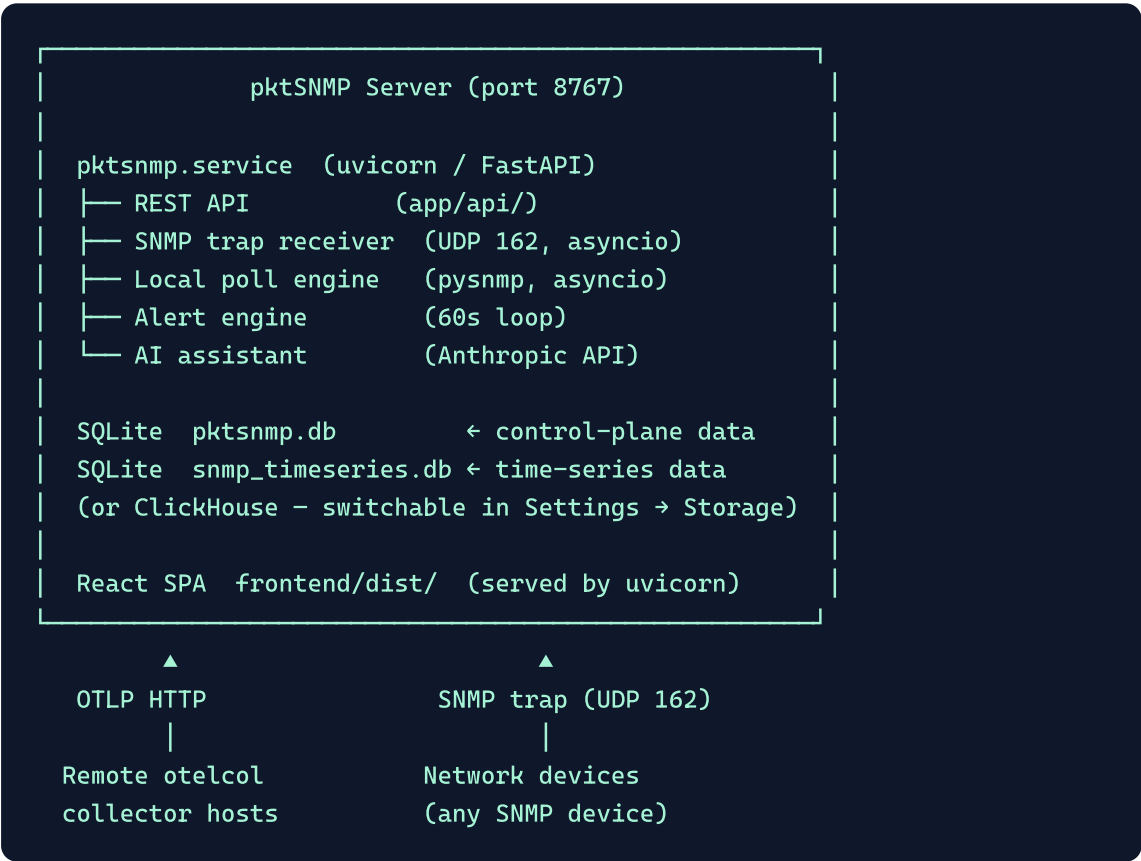


pktSNMP Administration Guide

Deployment, configuration, and maintenance reference for system administrators.

Architecture Overview

pktSNMP is a single systemd service combining a FastAPI backend, background workers, and a React SPA — all on one server with no required external dependencies beyond Python.



Layer	Technology
Backend	FastAPI + aiosqlite (Python 3.11+)
Frontend	React 18 + TypeScript + Tailwind CSS + Vite

Control-plane DB	SQLite pktsnmp.db — users, devices, collectors, alerts
Time-series store	SQLite snmp_timeseries.db (default) or ClickHouse
Auth	JWT + httpOnly refresh cookie + SAML 2.0
SNMP	pysnmp-lexstudio (v1/v2c/v3 traps and polling)
Service	systemd pktsnmp.service

Prerequisites

On the server: Python 3.11+, Node.js 20+ via NVM, Git.

Optional: ClickHouse (only if switching from SQLite storage backend).

Build the frontend on the server — never on Windows. The `node_modules` directory contains Linux-native binaries that won't cross-compile.

Fresh Install

1 Clone the repository on the server

```
ssh -i your-key.pem user@YOUR-SERVER-IP
cd /mnt/software
git clone git@github.com:your-org/pktsnmp.git pktsnmp
cd pktsnmp
```

2 Create a Python virtual environment

```
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

3 Create config.yaml

```
cp config.example.yaml config.yaml
# Edit: set secret_key (openssl rand -hex 32), db_path, log_file
```

4 Install and start the systemd service

```
sudo cp pktsnmp.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable pktsnmp
sudo systemctl start pktsnmp
```

5 Build the frontend

```
cd /mnt/software/pktsnmp/frontend
source ~/.nvm/nvm.sh && nvm use 20
npm ci && npm run build
sudo systemctl restart pktsnmp
```

6 First login

Browse to <http://YOUR-SERVER-IP:8767> . Log in with default admin credentials and change the password immediately.

Collector Setup

Local Collector (built-in)

Runs in-process. Enable via **Settings** → **SNMP** in the UI. Polls all devices assigned `collector_id=1` and listens on UDP 162 for traps. Requires `AmbientCapabilities=CAP_NET_BIND_SERVICE` (set in the service file).

Remote otelcol Collector

Deploy `otelcol-contrib` at each remote site. Add to the otelcol config:

```
exporters:
  otlphttp/pktsnmp:
    endpoint: "http://YOUR-SERVER-IP:8767/api/snmp/ingest/otlp"
    headers:
      Authorization: "Bearer YOUR_COLLECTOR_TOKEN"
    tls:
      insecure: true
    timeout: 30s
```

otelcol appends `/v1/metrics` to the endpoint automatically. The actual POST hits `/api/snmp/ingest/otlp/v1/metrics`. Both paths are registered in pktSNMP.

Metric naming convention (expected by the parser):

```
SNMP/<SITE>/<DEVICE>/<OID_LABEL>
# e.g.  SNMP/SITE-A/CORE-SW1/ifInOctets
```

Metrics not prefixed with `SNMP/` are ignored.

Device registration: Set the `otelcol_label` field on each device record to match the metric path prefix (e.g., `SITE-A/CORE-SW1`). When matched, the device's `last_seen` and status are updated automatically on each ingest batch.

Configuration Reference

Key	Default	Description
host	0.0.0.0	Bind address
port	8767	HTTP port
workers	2	uvicorn workers
secret_key	—	JWT signing secret — must change
db_path	/mnt/software/pktsnmp/pktsnmp.db	Control-plane SQLite path
clickhouse_host	localhost	ClickHouse host (if used)
log_level	info	debug / info / warning / error
log_file	/mnt/software/logs/pktsnmp.log	Log output path

Service Management

```
# Status
sudo systemctl status pktsnmp

# Live logs
sudo journalctl -u pktsnmp -f

# Restart / stop / start
sudo systemctl restart pktsnmp
sudo systemctl stop pktsnmp
sudo systemctl start pktsnmp
```

Auth & SAML

Local Auth

JWT access tokens expire in 15 minutes; httpOnly refresh cookies last 7 days. Change passwords via Settings → Users or the key icon in the sidebar.

SAML 2.0 (Okta or compatible IdP)

Configure in **Settings** → **Auth**:

1. Enter IdP Entity ID, SSO URL, and certificate.
2. Create a SAML app in your IdP:
 - ACS URL: `https://YOUR-FQDN:8767/api/auth/saml/callback`
 - Audience URI: `https://YOUR-FQDN:8767/api/auth/saml/metadata`
 - Name ID format: EmailAddress
3. Add attribute statement: Name = `role` , Value = `admin` / `analyst` / `viewer` .

HTTPS is required for SAML. The ACS URL must match your TLS certificate hostname. HTTP is not supported for the SAML callback.

Storage Backends

Backend	Best for	Notes
SQLite (default)	Most deployments — zero dependency, easy backup	Automatic — no setup needed
ClickHouse	High-volume environments, long-term retention at scale	Requires running ClickHouse server; set host/database in config.yaml

Switch in **Settings** → **Storage**. Restart the service after switching. Data is **not migrated** between backends.

Backup & Restore

Trigger backup via API:

```
curl -X POST http://YOUR-SERVER-IP:8767/api/system/backup \
-H "Authorization: Bearer YOUR_TOKEN"
```

Manual backup:

```
sudo systemctl stop pktsnmp
cp /mnt/software/pktsnmp/pktsnmp.db \
  /mnt/software/pktsnmp_backups/pktsnmp_$(date +%Y%m%d_%H%M%S).db
cp /mnt/software/pktsnmp/snmp_timeseries.db \
  /mnt/software/pktsnmp_backups/snmp_ts_$(date +%Y%m%d_%H%M%S).db
sudo systemctl start pktsnmp
```

Restore:

```
sudo systemctl stop pktsnmp
cp /mnt/software/pktsnmp_backups/pktsnmp_<ts>.db \
  /mnt/software/pktsnmp/pktsnmp.db
cp /mnt/software/pktsnmp_backups/snmp_ts_<ts>.db \
```

```
/mnt/software/pktsnmp/snmp_timeseries.db  
sudo systemctl start pktsnmp
```

Upgrading

```
cd /mnt/software/pktsnmp  
git pull  
source venv/bin/activate  
pip install -r requirements.txt  
cd frontend && source ~/.nvm/nvm.sh && nvm use 20  
npm ci && npm run build  
sudo systemctl restart pktsnmp
```

Migrations run automatically at startup — no manual schema steps needed.

Troubleshooting

Symptom	What to check
Service won't start	<code>journalctl -u pktsnmp -n 50</code> ; verify config.yaml paths and secret_key
Port 162 bind fails	Verify <code>AmbientCapabilities=CAP_NET_BIND_SERVICE</code> in service file; reload daemon and restart
No data from remote collector	Check otelcol logs on collector host; verify bearer token; confirm <code>otelcol_label</code> matches metric path prefix
401 on /ingest/otlp	Token mismatch — rotate token in Collectors page and update otelcol config
Collector status "unknown"	otelcol hasn't pushed data yet; verify otelcol is running and endpoint is reachable
devices.last_seen not updating	Verify <code>otelcol_label</code> on device matches metric path prefix; check ingest returns 202
Frontend blank / 404	Frontend build didn't complete; verify <code>frontend/dist/</code> exists; rebuild
Alert fires but dot stays green	Alert engine evaluates every 60s — wait one cycle; if persists, check <code>devices.status</code> in SQLite
Storage backend wrong	<code>storage_backend</code> setting takes effect on next restart; restart the service after switching
ClickHouse not found	Verify ClickHouse is running: <code>systemctl status clickhouse-server</code> ; check credentials in config.yaml